

No Universal Bohr Radius for Arbitrary Operator-Valued Biholomorphic Curves

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Status. Candidate scoped counterexample, likely valid, for human review.

1 Source question

Bhowmik and Das consider an operator-valued holomorphic function $f : \mathbb{D} \rightarrow \mathcal{B}(\mathcal{H})$ and a subordinate function

$$g(z) = \sum_{k=0}^{\infty} B_k z^k \in S(f).$$

Their final Remark (ii) asks whether there is a universal $r_0 > 0$ such that for every biholomorphic f , every $g \in S(f)$, and every $0 \leq r \leq r_0$,

$$\sum_{k=1}^{\infty} \|B_k\| r^k \leq \liminf_{|z| \rightarrow 1^-} \|f(z) - f(0)\|. \quad (1)$$

They suggest that the answer might be negative, perhaps even when f is convex [1, final Remark (ii), PDF p. 13].

Remarks. We end the article with the following observations:

- (i) It is immediately seen that for complex valued function f , part (i) of the Theorem [4](#) converts to the Bohr inequality for $S(f)$ where f is a normalized starlike univalent function. Again, if f is a complex valued normalized starlike univalent function defined on $\mathbb{D}$, the right hand side of the inequality in part (ii) is converted to $1/4$ which is known to be less or equal to $d(f(0), \partial\Omega)$, $\partial\Omega$ being the boundary of $\Omega = f(\mathbb{D})$, and thereby showing that part (ii) can also be considered as an operator valued analogue of the Bohr phenomenon for $S(f)$. We note that the scalar valued result is a direct consequence of [\[1\]](#) Theorem 1].
- (ii) In view of Theorem [3](#) and Theorem [4](#), it is a natural question to ask if the inequality [\(2.26\)](#) holds for $|z| = r \leq r_0$ for some $r_0 > 0$, where f is any function in the entire family of biholomorphic functions from $\mathbb{D}$ to $\mathcal{B}(\mathcal{H})$ and $g \in S(f)$. The Bohr radius $1/(1 + 2\|A_1\|\|A_1^{-1}\|)$ determined in the first part of the Theorem [3](#) is not bounded below by a positive constant if we allow A_1 to be any invertible operator from $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ varying on the family of complex Hilbert spaces. Therefore we remark that the answer of the aforesaid question could possibly be negative, even for f being convex biholomorphic, and that this can be an interesting problem for future research. However, similar problem for Banach space valued holomorphic functions in $\mathbb{D}$ has already been settled (cf. [\[7\]](#) Theorem 1.2)], where the notion of the Bohr inequality is analogous to [\(1.1\)](#).

2 Convention needed to read the question

The paper's printed definition says that $f(\mathbb{D})$ is a domain in the ambient space $\mathcal{B}(\mathcal{H})$. If “domain” has its standard meaning of an ambient open set, this is incompatible with every non-scalar example: the derivative of a map from $\mathbb{C}$ has rank at most one, so it cannot have a holomorphic local inverse onto an open subset of a space of dimension greater than one. Thus, read literally, the operator-valued part of the final question is empty.

The paper operates with the natural curve-level convention instead: f is holomorphic and injective, and its inverse on the one-dimensional image is holomorphic. Two internal signals make this intent concrete. The final question treats $A_1 = f'(0)$ as an invertible member of $\mathcal{B}(\mathcal{H})$, not as an isomorphism $\mathbb{C} \rightarrow \mathcal{B}(\mathcal{H})$. More decisively, the downloadable arXiv TeX contains a commented-out explanatory paragraph that calls $z \mapsto \text{diag}(z, t(z))$ biholomorphic for arbitrary holomorphic t , in order to observe that its higher coefficients can be arbitrarily large.

The following theorem answers the question under that operative convention. It does not claim to satisfy the incompatible ambient-open clause.

3 Claimed contribution

Theorem 1. *Under the source paper's operative curve convention, there is no universal $r_0 > 0$ for which (1) holds for every operator-valued biholomorphic f and every $g \in S(f)$. A counterexample exists in $\mathcal{B}(\mathbb{C}^2)$ for every proposed positive radius.*

4 Proof intuition

Keep one diagonal coordinate equal to z . This permanently records the parameter, so injectivity and a holomorphic inverse are automatic. Use the other coordinate to add two arbitrarily large, cancelling Taylor coefficients. Their norms add on the left of (1), but the cancellation makes that coordinate return to the value one at the boundary. The first coordinate simultaneously forces the boundary liminf to be exactly one. Thus the coefficient side can be made larger than the geometric side at any prescribed positive radius.

5 Proof

Let $r_0 > 0$ be arbitrary and put

$$r = \min\{r_0/2, 1/2\}.$$

Then $0 < r \leq r_0$. Choose a real number

$$M > \frac{1-r}{r^2(1+r)} \quad (2)$$

and take $\mathcal{H} = \mathbb{C}^2$. Define

$$f_M(z) = \begin{pmatrix} z & 0 \\ 0 & z + Mz^2 - Mz^3 \end{pmatrix}, \quad z \in \mathbb{D}, \quad (3)$$

and let $g = f_M$.

The map f_M is a holomorphic polynomial. Its first diagonal coordinate is z , so it is injective. Its inverse on $f_M(\mathbb{D})$ is the restriction of the bounded complex-linear coordinate functional

$$\pi_{11} : \mathcal{B}(\mathbb{C}^2) \longrightarrow \mathbb{C}, \quad \pi_{11}(T) = T_{11}.$$

Consequently f_M is biholomorphic under the operative convention. Taking the identity Schwarz function $\phi(z) = z$ gives $g = f_M \circ \phi$, hence $g \in S(f_M)$.

Since $f_M(0) = 0$, the nonzero Taylor coefficients of g are

$$B_1 = I_2, \quad B_2 = \text{diag}(0, M), \quad B_3 = \text{diag}(0, -M).$$

Their operator norms are $1, M, M$, respectively. Therefore the left-hand side of (1) at the chosen radius is

$$\begin{aligned} \sum_{k=1}^{\infty} \|B_k\| r^k &= r + Mr^2 + Mr^3 \\ &= r + Mr^2(1+r) > 1, \end{aligned}$$

where the last inequality is exactly (2).

It remains to compute the right-hand side. For every $z \in \mathbb{D}$,

$$\|f_M(z)\| = \max\{|z|, |z + Mz^2 - Mz^3|\} \geq |z|.$$

It follows that

$$\liminf_{|z| \rightarrow 1^-} \|f_M(z)\| \geq 1.$$

On the other hand, along positive real numbers $z = \rho \uparrow 1$,

$$\rho + M\rho^2 - M\rho^3 = \rho + M\rho^2(1 - \rho) \longrightarrow 1,$$

while the first diagonal coordinate ρ also tends to one. Hence

$$\lim_{\rho \uparrow 1} \|f_M(\rho)\| = 1,$$

and therefore

$$\liminf_{|z| \rightarrow 1^-} \|f_M(z) - f_M(0)\| = 1. \quad (4)$$

The strict coefficient estimate and (4) contradict (1) at $r \leq r_0$. Since $r_0 > 0$ was arbitrary, no universal positive radius exists. $\square$

6 Verification and limitations

The script `code/verify_counterexample.py` checks the coefficient identities, the choice of M , and the strict inequality with exact rational arithmetic for representative radii. The proof itself is symbolic and uses no numerical approximation.

The construction settles the unrestricted question only under the operative curve convention. It is not convex. Accordingly it does not settle the stronger suggestion that failure may persist for convex biholomorphic maps. Under the printed ambient-open definition, the non-scalar class is empty instead; that is a defect in the formulation, not an additional conclusion of the counterexample.

7 Bounded novelty check

The search on 27 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; exact searches for arXiv:1904.00351, its title, and phrases from the final question; the arXiv source and Cambridge publisher page; and the later operator-valued Bohr papers surfaced by those searches. No explicit proof or counterexample to this unrestricted question was found. This supports, but cannot certify, novelty.

Human review recommendation. Ask an expert in holomorphic mappings in Banach spaces or operator-valued Bohr phenomena to confirm the source convention. Once the operative convention is accepted, the mathematical calculation is elementary and exact.

References

- [1] B. Bhowmik and N. Das, *Bohr phenomenon for operator valued functions*, Proc. Edinburgh Math. Soc. **64** (2021), 72–86; arXiv:1904.00351, <https://arxiv.org/abs/1904.00351>; <https://doi.org/10.1017/S0013091520000395>.